

Table 1: One-pass residual readout vs. fairly-conditioned autoregressive loop (accuracy). Readout = small MLP on the frozen final residual (4-seed vote); loop = next-token log-prob over the closed answer set, full context, balanced few-shot. Δ = readout – loop.k, paired McNemar on identical rows (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). All runs batch 8; Qwen3 0.6B/4B on H200, 8B on B200, DeepSeek-V4-Flash on B300. ^a BoolQ loops: Qwen at pad 8192 with best $k \in \{0, \dots, 64\}$, DSV4 at pad 2048, $k=8$. ^b RuleTaker n2k: readout scored on the loop’s exact 400 rows. ^c ARC-Challenge: full test (1165 rows), $k=8$, pad 2048.

Task	Model	Readout	loop.0	loop.k	Δ	p (McNemar)
BoolQ ^a	Qwen3-0.6B	0.753	0.631	0.715	+0.038***	2.4e−05
	Qwen3-4B	0.862	0.854	0.869	−0.007	0.22
	Qwen3-8B	0.879	0.862	0.886	−0.007	0.24
	DeepSeek-V4-Flash	0.896	0.888	0.906	−0.009	0.39
RuleTaker n2k ^b	Qwen3-0.6B	0.638	0.600	0.545	+0.093**	3.7e−03
	Qwen3-4B	0.738	0.653	0.698	+0.040	0.20
	Qwen3-8B	0.753	0.675	0.730	+0.023	0.45
	DeepSeek-V4-Flash	0.763	0.605	0.838	−0.074**	1e−03
ARC-Challenge ^c	Qwen3-0.6B	0.492	0.502	0.597	−0.106***	3.2e−11
	Qwen3-4B	0.837	0.826	0.882	−0.046***	2.9e−06
	Qwen3-8B	0.908	0.902	0.913	−0.005	0.57
	DeepSeek-V4-Flash	0.951	0.961	0.959	−0.007	0.23

Table 2: Readout probe-placement sweeps (12 cells, all batch 8): best mid-depth tap vs the final-layer readout, paired McNemar on identical rows. Mid-depth taps win significantly at 6/12 cells and never lose significantly; the optimum sits at 64–86% depth on BoolQ/RuleTaker and drifts to the final layer on ARC at $\geq 8B$.

Task	Model	Best tap	Depth	mlp (best)	mlp (final)	Δ	p
BoolQ	Qwen3-0.6B	L18	64%	0.761	0.746	+0.010	0.18
	Qwen3-4B	L24	67%	0.868	0.856	+0.011*	0.038
	Qwen3-8B	L30	83%	0.889	0.878	+0.011*	0.016
	DeepSeek-V4-Flash	L36	86%	0.900	0.896	+0.005	0.20
RuleTaker n2k	Qwen3-0.6B	L18	64%	0.688	0.650	+0.036*	0.020
	Qwen3-4B	L24	67%	0.781	0.744	+0.038**	2.8e−03
	Qwen3-8B	L24	67%	0.778	0.758	+0.025*	0.040
	DeepSeek-V4-Flash	L29	67%	0.776	0.765	+0.021	0.057
ARC-Challenge	Qwen3-0.6B	L26	93%	0.500	0.492	+0.013	0.082
	Qwen3-4B	L29	81%	0.856	0.841	+0.022***	2.2e−04
	Qwen3-8B	L35	97%	0.909	0.909	+0.001	1.00
	DeepSeek-V4-Flash	L42	98%	0.951	0.951	−0.005	0.29